No regular icosahedron has a rational point on every face: a proof checked in Lean

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Abstract

We prove that no regular icosahedron in $\mathbb{R}^3$ has a point with rational coordinates on each of its twenty closed faces, in every position, size and orientation, reflections included. This answers a question of I. Bogdanov on MathOverflow. A field automorphism of $\mathbb{C}$ that sends $\sqrt{5}$ to $-\sqrt{5}$ reduces the problem to the following statement: for every complex orthogonal 3×3 matrix T some real linear combination of $\operatorname{Re} T$ and $\operatorname{Im} T$ separates ten intervals determined by the faces. We prove this statement by a computer-assisted case analysis over rational parametrizations of rotation frames, in which every inequality is an exact polynomial bound over $\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$; no floating-point arithmetic and no Lipschitz estimates are used. The whole argument, including the 48 certificates, is formalized in the Lean 4 proof assistant with Mathlib and checked by the Lean kernel; the main theorem depends only on the standard axioms. The mathematical argument and the formalization were produced by AI systems (OpenAI Codex and Anthropic Claude) under the direction of the author; see the statement at the end of the paper.

1 Introduction

Can a regular icosahedron contain a rational point on each of its faces? The question was asked by Ilya Bogdanov on MathOverflow in August 2023 [1] and has remained without an answer there. For the other Platonic solids the answer is positive and easy: regular tetrahedra, cubes and octahedra may have all vertices rational, and eight vertices of a regular dodecahedron form a cube whose edges are diagonals of the twelve pentagonal faces [1]. A regular icosahedron cannot have all vertices rational, since there is no regular pentagon in an integer lattice of any dimension; but this alone says nothing about one rational point per face.

Theorem 1.1. *Let $I \subset \mathbb{R}^3$ be a regular icosahedron. Then at least one of the twenty closed triangular faces of I contains no point of $\mathbb{Q}^3$.*

Points on edges and vertices are allowed; a face is the closed triangle. Closedness matters: in the comments to [1] J. Rouse gives a regular icosahedron and six rational points that together lie on all twenty face *planes*, so that the analogue of the question for planes has a positive answer. The planar analogue of the theorem is known: there is no regular pentagon with a rational point on each edge [2]. That proof is algebraic and works even for rational points on the lines through the edges; M. Bolan extended it to all regular n-gons, which have a rational point on each edge only for $n = 3, 4, 8$ [2]. By Rouse's example, no argument of this kind can work for the icosahedron: a proof has to use the boundaries of the faces.

Our proof has two parts. The first part (Section 2) is algebraic. The planes of the faces are defined over $K = \mathbb{Q}(\sqrt{5})$; applying an automorphism τ of $\mathbb{C}$ with $\tau(\sqrt{5}) = -\sqrt{5}$ to the equations

of the faces and the rational points turns a hypothetical solution into a complex orthogonal matrix $T \in SO(3,\mathbb{C})$ such that, for every real α, β, ten real intervals built from the matrix $\alpha \operatorname{Re} T + \beta \operatorname{Im} T$ and the ten pairs of opposite faces have a common point. The second part (Sections 3 and 4) shows that this never happens: for every $T \in SO(3,\mathbb{C})$ some combination separates the intervals. Linear algebra reduces this to a statement about a compact parameter space, which is then checked on finitely many cells by exact arithmetic.

The whole proof is formalized in Lean 4 [3] with Mathlib [4] and checked by the Lean kernel (Section 5). The computational part is carried out by the kernel itself through `decide +kernel` on 48 certificates; `native_decide` and other ways to trust compiled code are not used. The only parts that must be trusted are the Lean kernel and the definitions in the statement: 58 lines that define the canonical icosahedron, its faces and rational points.

2 Reduction to complex orthogonal matrices

Let $\varphi = (1+\sqrt{5})/2$, $\varphi' = 1-\varphi = (1-\sqrt{5})/2$, $K = \mathbb{Q}(\sqrt{5})$, and let $x \mapsto x'$ denote the nontrivial automorphism of K. The canonical icosahedron I_0 has the twelve vertices obtained from $(0, \pm 1, \pm\varphi)$ by cyclic permutations of the coordinates, and edge length 2. Every regular icosahedron is the image of I_0 under $x \mapsto t + sRx$ with $t \in \mathbb{R}^3$, $s > 0$ and R orthogonal; this is how Theorem 1.1 is stated in Lean.

The twenty faces $F_0, \dots, F_{19}$ of I_0 come in ten pairs of opposite faces; we fix representatives $F_0, \dots, F_9$, so that the face opposite to F_i is $-F_i$. Each face F_j lies in a plane $n_j \cdot x = \varphi^2$ with $n_j \in K^3$: the eight normals are $(\pm 1, \pm 1, \pm 1)$ and the twelve others are cyclic permutations of $(0, \pm\varphi, \pm(\varphi - 1))$. We write n_j' for the conjugate normal and v_{i0}, v_{i1}, v_{i2} for the vertices of F_i.

Lemma 2.1 (centering)**.** *Suppose that both $t + sRF_i$ and the opposite face $t + sR(-F_i)$ contain rational points. Then there is $m \in F_i$ with $sRm \in \mathbb{Q}^3$.*

Proof. If $t + sRx = p$ and $t - sRy = q$ with $x, y \in F_i$ and $p, q \in \mathbb{Q}^3$, then $m = (x+y)/2 \in F_i$ by convexity and $sRm = (p-q)/2$. □

Lemma 2.2. *There is a field automorphism τ of $\mathbb{C}$ with $\tau(\sqrt{5}) = -\sqrt{5}$, that is, $\tau(\varphi) = \varphi'$.*

Proof. Extend the conjugation of K to a purely transcendental extension of K in $\mathbb{C}$ generated by a transcendence basis, and then to its algebraic closure $\mathbb{C}$. □

For a real matrix A and $0 \le i \le 9$, $0 \le k \le 2$ put

$$f_{ik}(A) = n_i'^{\mathsf{T}} A\, v_{ik}, \qquad J_i(A) = \big[\min_k f_{ik}(A),\ \max_k f_{ik}(A)\big].$$

Since $v \mapsto n_i'^{\mathsf{T}} A v$ is linear, $J_i(A)$ is the set of its values on the triangle F_i. We say that A *separates* if $J_0(A), \dots, J_9(A)$ have no common point. Clearly, if A separates, then so does cA for every $c > 0$.

Proposition 2.3. *If every face of $t + sRI_0$ contains a rational point, then the matrix $T = (R^\tau)^{\mathsf{T}} R$, where τ is applied entrywise, satisfies $T^{\mathsf{T}} T = I$ and $\det T = 1$, and no matrix $\alpha \operatorname{Re} T + \beta \operatorname{Im} T$ with $\alpha, \beta \in \mathbb{R}$ separates.*

Proof. By Lemma 2.1 there are $m_i \in F_i$ with $p_i = sRm_i \in \mathbb{Q}^3$. Since $R^{\mathsf{T}} R = I$ and $n_i \cdot m_i = \varphi^2$, we have $(Rn_i) \cdot p_i = s\varphi^2$. Apply τ: it fixes p_i, maps the entries of n_i to those of n_i' and φ to φ', so $(R^\tau n_i') \cdot p_i = \tau(s)\varphi'^2$. Substituting $p_i = sRm_i$ gives

$$n_i'^{\mathsf{T}}\, T\, m_i = \kappa\, \varphi'^2, \qquad \kappa = \tau(s)/s \in \mathbb{C}, \qquad i = 0, \dots, 9.$$

Since n_i', m_i and φ'^2 are real, for $A = \alpha \operatorname{Re} T + \beta \operatorname{Im} T$ we get $n_i'^{\mathsf{T}} A m_i = (\alpha \operatorname{Re}\kappa + \beta \operatorname{Im}\kappa)\varphi'^2$, the same number for all i, and it lies in every $J_i(A)$. Finally, $T^{\mathsf{T}}T = R^{\mathsf{T}}\tau(RR^{\mathsf{T}})R = I$ and $\det T = \tau(\det R)\det R = (\det R)^2 = 1$, because $\det R = \pm 1$ is fixed by τ. □

Hence Theorem 1.1 follows from the next theorem, which is the main work.

Theorem 2.4 (separation). *For every $T \in M_3(\mathbb{C})$ with $T^{\mathsf{T}}T = I$ and $\det T = 1$ there are $\alpha, \beta \in \mathbb{R}$ such that $\alpha \operatorname{Re} T + \beta \operatorname{Im} T$ separates.*

3 Proof of the separation theorem

3.1 Chart frames

For $u, v \in \mathbb{R}$ let $Q(u,v)$ be the rotation matrix of the quaternion $1 + u\,\mathbf{i} + v\,\mathbf{j}$ without normalization:

$$Q(u,v) = \begin{pmatrix} 1+u^2-v^2 & 2uv & 2v \\ 2uv & 1-u^2+v^2 & -2u \\ -2v & 2u & 1-u^2-v^2 \end{pmatrix}, \qquad \mu(u,v) = 1+u^2+v^2.$$

Then $Q^{\mathsf{T}}Q = QQ^{\mathsf{T}} = \mu^2 I$ and $\det Q = \mu^3$, so $Q/\mu \in SO(3)$. Let $P_0, \dots, P_5$ be the signed permutation matrices with columns $e_i, \sigma e_j, \sigma e_k$, where (i,j,k) is a cyclic shift of $(1,2,3)$ and $\sigma = \pm 1$; they satisfy $P_c^{\mathsf{T}}P_c = I$, $\det P_c = 1$. The *chart frame* is $\operatorname{Fr}_c(u,v) = P_c Q(u,v)$. Its third column is $P_c(2v, -2u, 1-u^2-v^2)^{\mathsf{T}}$, which is μ times a unit vector.

Lemma 3.1 (six charts). *For every unit vector $d \in \mathbb{R}^3$ there are $c \in \{0,\dots,5\}$ and u, v with $|u|, |v| \le 1/2$ and $\operatorname{Fr}_c(u,v)e_3 = \mu(u,v)\,d$.*

Proof. Choose c so that $P_c^{\mathsf{T}}d = (a_1, a_2, a_3)$ has $a_3 \ge |a_1|, |a_2|$ (the coordinate of d of largest absolute value, with its sign). Put $u = -a_2/(1+a_3)$, $v = a_1/(1+a_3)$. From $a_1^2 + a_2^2 = (1-a_3)(1+a_3)$ one gets $\mu = 2/(1+a_3)$ and $(2v, -2u, 1-u^2-v^2) = \mu(a_1, a_2, a_3)$; moreover $|u|, |v| \le a_3/(1+a_3) \le 1/2$. □

Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and, for real numbers,

$$M_r(\sigma, t) = \operatorname{diag}\big(\sigma((1-t^2)I_2 + 2tJ),\ 1+t^2\big), \qquad M_c(x,y,z) = \operatorname{diag}\big(xI_2 + yJ,\ z\big).$$

The computational part of the proof is the following theorem; its certificates are described in Section 4.

Theorem 3.2 (certificates). (R) *For every chart c, every sign $\sigma = \pm 1$ and all $|u|, |v| \le 1/2$, $|t| \le 1$ the matrix $\operatorname{Fr}_c(u,v) M_r(\sigma, t)$ separates.*

(C) *For every pair of charts c_1, c_2 and all $u, v, u', v' \in [-1/2, 1/2]$ there is a triple (X, Y, D) from Table 1, depending on (u, v, u', v'), such that for all $z \in [-1, 1]$ the matrix $\operatorname{Fr}_{c_1}(u,v)\, M_c(X, Y, Dz)\, \operatorname{Fr}_{c_2}(u',v')^{\mathsf{T}}$ separates.*

3.2 The real case

Write $X = \operatorname{Re} T$, $Y = \operatorname{Im} T$. From $T^{\mathsf{T}}T = I$:

$$X^{\mathsf{T}}X = I + Y^{\mathsf{T}}Y, \qquad X^{\mathsf{T}}Y + Y^{\mathsf{T}}X = 0. \tag{1}$$

X	1	99	12	91	119	161	5	20	0
Y	0	20	5	60	120	240	12	99	1
D	1	101	13	109	169	289	13	101	1

Table 1: The sixteen phases $(\pm X, Y, D)$, $X^2+Y^2=D^2$: the nine listed triples and the seven triples $(-X, Y, D)$ with $X \neq 0, 1$. They are directions in the upper half plane.

Lemma 3.3 (frame decomposition)**.** *Let $T^{\mathsf T}T = I$, $\det T = 1$, let $b \in \mathbb{R}^3$ satisfy $Yb = 0$, and let the chart frames $F_1 = \mathrm{Fr}_{c_1}(u, v)$, $F_2 = \mathrm{Fr}_{c_2}(u', v')$ satisfy $F_1 e_3 = \mu\, Xb$ and $F_2 e_3 = \mu' b$, where $\mu = \mu(u, v)$, $\mu' = \mu(u', v')$. Then there are $p, r \in \mathbb{C}$ with $p^2 + r^2 = 1$ and*

$$F_1^{\mathsf T}\, T\, F_2 = \mu\mu'\ \mathrm{diag}(pI_2 + rJ,\ 1).$$

Proof. Let $M = F_1^{\mathsf T} T F_2$ and $k = \mu\mu'$. Then $M^{\mathsf T}M = F_2^{\mathsf T}T^{\mathsf T}(F_1F_1^{\mathsf T})TF_2 = k^2 I$ and $\det M = k^3$. Since $Yb = 0$, $TF_2e_3 = \mu' Tb = \mu' Xb = (\mu'/\mu)F_1e_3$, hence $Me_3 = (\mu'/\mu)F_1^{\mathsf T}F_1e_3 = ke_3$. Then $M^{\mathsf T}e_3 = k^{-1}M^{\mathsf T}Me_3 = ke_3$, so the third row and column of M are $(0, 0, k)$. The upper left block P satisfies $P^{\mathsf T}P = k^2 I_2$ and $k \det P = k^3$; for $P = \left(\begin{smallmatrix} a & b' \\ c & d \end{smallmatrix}\right)$ this gives $d = a$, $b' = -c$, and $p = a/k$, $r = c/k$. □

Lemma 3.4. *If $\mathrm{Im}\, T = 0$, then $\mathrm{Re}\, T = \gamma\ \mathrm{Fr}_c(u, v) M_r(\sigma, t)$ for some $\gamma > 0$, chart c, sign σ and $|u|, |v| \le 1/2$, $|t| \le 1$.*

Proof. By (1) X is orthogonal, so Xe_3 is a unit vector. Take c, u, v from Lemma 3.1 for $d = Xe_3$ and apply Lemma 3.3 with $b = e_3$ and $F_2 = \mathrm{Fr}_0(0, 0) = I$. Since $T = X$ is real, so are p, r, and $p^2 + r^2 = 1$. If $p \ge 0$ put $\sigma = 1$, $t = r/(1 + p)$; otherwise $\sigma = -1$, $t = r/(p - 1)$. In both cases $|t| \le 1$ and $M_r(\sigma, t) = (1 + t^2)\,\mathrm{diag}(pI_2 + rJ, 1)$ (the half-angle substitution). Since $F_1F_1^{\mathsf T} = \mu^2 I$, we get $X = \mu^{-2}F_1(F_1^{\mathsf T}X) = (\mu(1 + t^2))^{-1}F_1M_r(\sigma, t)$. □

3.3 The non-real case

Lemma 3.5. *There is a unit vector $b \in \mathbb{R}^3$ with $Yb = 0$; for it $\|Xb\| = 1$.*

Proof. By (1), $\|Xw\|^2 = \|w\|^2 + \|Yw\|^2$ for every w, so X is invertible. From $X^{\mathsf T}Y = -Y^{\mathsf T}X$ we get $\det X \det Y = -\det Y \det X$, hence $\det Y = 0$. Take b in the kernel of Y; then $\|Xb\|^2 = \|b\|^2 = 1$. □

Proposition 3.6. *Let $\mathrm{Im}\, T \neq 0$. There are charts c_1, c_2 and $u, v, u', v' \in [-1/2, 1/2]$ such that for every (X_0, Y_0, D) with $D > 0$ and $X_0^2 + Y_0^2 = D^2$ there are $\alpha, \beta \in \mathbb{R}$, $\gamma > 0$ and $z \in [-1, 1]$ with*

$$\alpha\, \mathrm{Re}\, T + \beta\, \mathrm{Im}\, T = \gamma\ \mathrm{Fr}_{c_1}(u, v)\, M_c(X_0, Y_0, Dz)\ \mathrm{Fr}_{c_2}(u', v')^{\mathsf T}.$$

Proof. Take b from Lemma 3.5, frames F_1 for Xb and F_2 for b from Lemma 3.1, and $p = p_1 + ip_2$, $r = r_1 + ir_2$ from Lemma 3.3. Taking real and imaginary parts,

$$F_1^{\mathsf T}XF_2 = \mu\mu' M_c(p_1, r_1, 1), \qquad F_1^{\mathsf T}YF_2 = \mu\mu' M_c(p_2, r_2, 0),$$

and $p^2 + r^2 = 1$ gives $a = p_1^2 + r_1^2 = 1 + b''$, $b'' = p_2^2 + r_2^2$, and $p_1p_2 + r_1r_2 = 0$. If $b'' = 0$, then $F_1^{\mathsf T}YF_2 = 0$ and $Y = (\mu\mu')^{-2}F_1(F_1^{\mathsf T}YF_2)F_2^{\mathsf T} = 0$; so $b'' > 0$ and $a \ge 1$. Put

$$\alpha = (X_0p_1 + Y_0r_1)\, b'', \qquad \beta = (X_0p_2 + Y_0r_2)\, a, \qquad \lambda = ab'', \qquad z = \alpha/(\lambda D).$$

Using $p_1p_2 + r_1r_2 = 0$ one checks $\alpha p_1 + \beta p_2 = \lambda X_0$ and $\alpha r_1 + \beta r_2 = \lambda Y_0$, so $F_1^{\mathsf T}(\alpha X + \beta Y)F_2 = \mu\mu'\lambda\, M_c(X_0, Y_0, Dz)$. By Cauchy–Schwarz $|X_0p_1 + Y_0r_1| \le D\sqrt{a} \le Da$, hence $|z| \le 1$. Multiplying by F_1 on the left and by $F_2^{\mathsf T}$ on the right and using $F_iF_i^{\mathsf T} = \mu_i^2 I$ gives the claim with $\gamma = \lambda/(\mu\mu')$. □

Proof of Theorem 2.4. If $\operatorname{Im} T = 0$, take $\alpha = 1$, $\beta = 0$: by Lemma 3.4 and Theorem 3.2(R), $\operatorname{Re} T$ is a positive multiple of a separating matrix. If $\operatorname{Im} T \neq 0$, take the charts and parameters of Proposition 3.6; Theorem 3.2(C) gives a phase (X_0, Y_0, D) that separates for all $z \in [-1, 1]$, and Proposition 3.6 with this phase gives α, β. □

The frames in Proposition 3.6 do not depend on the phase, while the phase chosen by Theorem 3.2(C) depends on the frames; this order of quantifiers is what the proof needs.

4 The certificates

4.1 Exact arithmetic

All checks are computations with natural numbers that the Lean kernel evaluates directly; its natural numbers are arbitrary precision and addition, multiplication and comparison are implemented by GMP. An element $a + b\varphi$ of $\mathbb{Z}[\varphi]$ is stored as four naturals (a^+, a^-, b^+, b^-). Its sign is decided exactly: with $w = 2a + b$, $a + b\varphi \geq 0$ iff $w + b\sqrt{5} \geq 0$, which is decided by comparing w^2 with $5b^2$ when w and b have opposite signs. The checker uses only `Nat.add`, `Nat.mul`, `Nat.sub`, `Nat.ble`, structures, `cond` and explicit recursion `Nat.rec`, so that `decide +kernel` can reduce it.

4.2 Cells and polynomial bounds

A cell of a chart square at level m and index j is $u = (2j + 1 - 2^m + s)/2^{m+1}$, $s \in [-1, 1]$; at level 0 it is the whole $[-1/2, 1/2]$. A t-interval at level m is $t = (2j + 1 - 2^m + s)/2^m$. With $N = 2^{m+1}$ the entries of $N^2 Q(u, v)$ are quadratic polynomials in the two cell coordinates with integer coefficients, and so are the entries of $2^{2m} M_r(\sigma, t)$ in the coordinate of t. Hence, for $A = \operatorname{Fr}_{c_1} M \operatorname{Fr}_{c_2}^{\mathsf{T}}$,

$$f_{ik}(A) = \left(\operatorname{Fr}_{c_1}^{\mathsf{T}} n_i'\right) \cdot \left(M \operatorname{Fr}_{c_2}^{\mathsf{T}} v_{ik}\right) = (\text{positive constant}) \cdot a_i \cdot w_{ik},$$

where a_i and w_{ik} are vectors of polynomials of degree at most 2 in two variables over $\mathbb{Z}[\varphi]$; similarly in the real case with $w_{ik} = M_r v_{ik}$, a polynomial in one variable. The positive constant does not affect separation. Write $a = a^0 + \tilde{a}$ and $w = w^0 + \tilde{w}$ (constant terms and the rest). On $[-1, 1]^2$ the polynomial $c_1 s + c_2 t + c_3 s^2 + c_4 st + c_5 t^2$ is at least $\min(c_3, 0) + \min(c_5, 0) - |c_1| - |c_2| - |c_4|$, and $|\tilde{a} \cdot \tilde{w}| \leq \sum_l \|\tilde{a}_l\|_1 \|\tilde{w}_l\|_1$, where $\|\cdot\|_1$ is the sum of the absolute values of the coefficients. This gives an exact lower bound of $f_{i_1 k} - f_{i_2 k'}$ on the whole cell. A pair (i_1, i_2) *separates the cell* if these nine lower bounds, $k, k' \in \{0, 1, 2\}$, are positive; then $J_{i_1}(A)$ lies to the right of $J_{i_2}(A)$ for every A of the cell, and A separates. The soundness of this bound is proved in Lean.

4.3 The real case: search in the kernel

For (R) no certificate data is needed. The function `rsearch` takes a cell, chooses at its centre i_1 maximizing $\min_k f_{ik}$ and i_2 minimizing $\max_k f_{ik}$, and checks whether (i_1, i_2) separates the cell; if not, it splits the (u, v) square into four or the t-interval into two (the coarser one) and recurses, with a recursion bound of 40. The kernel runs this search for the six charts and two signs, twelve runs in total, and the results are the twelve theorems `rsearch c` σ `40` $\langle 0, 0, 0, 0, 0 \rangle$ `= true`.

4.4 The complex case: certificate trees

For (C) the search over (u, v, u', v') is guided by a tree, stored as one natural number whose hexadecimal digits are read from the least significant end: 0 means a split of the (u, v) or the (u', v') square (the coarser one) into four; $1\, n\, i_1\, i_2$ is a leaf where phase n and pair (i_1, i_2) separate the cell

at $z = -1$ and at $z = 1$; $2\,n\,i_1\,i_2\,j_1\,j_2$ is a leaf where (i_1, i_2) separates at $z = -1$ and $z = 0$ and (j_1, j_2) at $z = 0$ and $z = 1$. The forms f_{ik} are affine in z, so separation by a fixed pair at both ends of a z-interval implies it on the whole interval. The trees are produced by a generator (`Gen.lean`) that runs the same functions compiled and checks its output with them; the generator is not part of the proof. Table 2 gives the sizes.

pairs of charts	36
leaves with one pair	14 472
leaves with two pairs	4 896
splits	6 444
maximal total level $m_1 + m_2$	6
hexadecimal digits in all trees	93 708

Table 2: The certificate trees of the complex case.

4.5 Resources

On an Intel Core i5-12500 with 32 GB of memory the kernel checks one real certificate in 0.5–2 minutes using up to 7 GB, and one complex certificate in 1.5–3 minutes using 8–13 GB. Checking the complex certificates one at a time, all 48 take about 1.5 hours. Building them with the default parallelism of Lake exhausts 32 GB of memory, so the repository contains scripts that build them sequentially.

5 The formalization

The Lean project [5] uses Lean 4 v4.34.0-rc2 and Mathlib v4.34.0-rc2. Table 3 lists the files. The main theorem reads

```
theorem no_rational_point_on_every_face (t : Fin 3 → ℝ) (s : ℝ) (hs : 0 < s)
    (R : Matrix (Fin 3) (Fin 3) ℝ) (hR : R ∈ Matrix.orthogonalGroup (Fin 3) ℝ) :
    ¬ ∀ j : Fin 20, HasRatPoint t s R j
```

and its meaning depends only on the definitions in `Defs.lean`:

```
noncomputable def φ : ℝ := (1 + Real.sqrt 5) / 2
noncomputable def ev (x : ℤ × ℤ) : ℝ := x.1 + x.2 * φ
noncomputable def vertex (i : Fin 12) : Fin 3 → ℝ := fun c => ev (vtxCode i c)
noncomputable def face (j : Fin 20) : Set (Fin 3 → ℝ) :=
  convexHull ℝ (Set.range fun k => vertex (faceIdx j k))
def IsRat (x : Fin 3 → ℝ) : Prop := ∃ q : Fin 3 → ℚ, ∀ c, x c = q c
def HasRatPoint (t : Fin 3 → ℝ) (s : ℝ) (R : Matrix (Fin 3) (Fin 3) ℝ)
    (j : Fin 20) : Prop :=
  ∃ x ∈ face j, IsRat (t + s • (R *ᵥ x))
```

where `vtxCode` lists the twelve vertices $(0, \pm 1, \pm\varphi)$, $(\pm 1, \pm\varphi, 0)$, $(\pm\varphi, 0, \pm 1)$ as pairs (a, b) for $a + b\varphi$, and `faceIdx` lists the twenty faces as triples of vertex indices. The orthogonal group includes reflections. A short script in the repository checks that `faceIdx` consists exactly of the twenty triples of pairwise adjacent vertices, that is, of the faces. The command `#print axioms` gives

```
'Icosahedron.no_rational_point_on_every_face' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

file	lines	content
`Defs`	58	the statement: icosahedron, faces, rational points
`Geometry`	132	face planes, opposite faces, conjugate normals
`Embedding`	88	Lemma 2.2
`Reduction`	165	Lemma 2.1, Proposition 2.3
`KCheck`	487	the checker: $\mathbb{Z}[\varphi]$, polynomials, searches, phases
`KSound`	349	soundness of the arithmetic and of the cell bound
`KGeom`	535	chart frames; soundness of both searches
`Frames`	646	Lemmas 3.1–3.5, Proposition 3.6
`CertR/*`, `CertC/*`		48 certificates (`decide +kernel`), 110 KB
`CertAll`	131	collects the certificates
`Separation`	38	Theorem 2.4
`Main`	21	Theorem 1.1

Table 3: Files of the Lean library `Icosahedron`.

Use of AI tools

The author posed the problem, directed the work and checked the results. The mathematical argument (the reduction of Section 2 and the decomposition of Section 3) was developed by OpenAI Codex, and the Lean formalization, including the kernel-checkable certificates, their generator and the final form of the decomposition, was developed by Anthropic Claude, both under the direction of the author; the text of this paper was drafted with Anthropic Claude. The author takes full responsibility for the content. The correctness of Theorem 1.1 does not depend on trust in these systems: the proof is checked by the Lean kernel, and what must be trusted is the kernel and the 58 lines of definitions quoted in Section 5.

References

[1] I. Bogdanov, *Can a regular icosahedron contain a rational point on each face?*, MathOverflow, question 452847, 2023. https://mathoverflow.net/questions/452847

[2] *Is there a regular pentagon with a rational point on each edge?*, MathOverflow, question 459110 (asked by A. Ustinov), 2023; answers by uranix and M. Bolan, 2024. https://mathoverflow.net/questions/459110

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[4] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[5] M. Levit, *No regular icosahedron has a rational point on every face: Lean 4 formalization*, 2026. https://github.com/MikhailLevit1987/icosahedron-rational-faces